# Hepatitis B Core Antibody, IgM

## Test Details

**METHODOLOGY**

Immunochemiluminometric assay (ICMA)

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**RESULT TURNAROUND TIME**

1 - 2 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

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## Use

This assay may be used in combination with other hepatitis B virus (HBV) marker assays to define the clinical status of known HBV infected patients or can be combined with other HBV, HAV (hepatitis A virus), and HCV (hepatitis C virus) assays for the diagnosis of patients presenting with symptoms of acute viral hepatitis.

## Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

## Limitations

Assay performance characteristics have not been established for immunocompromised or immunosuppressed patients, cord blood, or patients less than 2 years of age.

Heterophilic antibodies in human serum can react with reagent immunoglobulins, interfering with in vitro immunoassays. Patients routinely exposed to animals or to animal serum products can be prone to this interfence and anomalous values may be observed. Additional information may be required for diagnosis.

A reactive anti-HBc IgM result does not exclude co-infection by another hepatitis virus.

Assay performance characteristics have not been established when the ADVIA Centaur HBc IgM assay is used in conjunction with other manufacturers' assays for specific HBV serological markers.

## Custom Additional Information

Anti-HBc IgM increases rapidly, peaks during the acute infection stage of HBV infection, and then falls to a relatively low level as the patient recovers or becomes a chronic carrier. Anti-HBc IgM is useful in the diagnosis of acute HBV infection even when HBsAg concentrations are below the sensitivity of the diagnostic assay. The presence of anti-HBc IgM and anti-HBc IgG is characteristic of acute infection, while the presence of anti-HBc IgG without anti-HBc IgM is characteristic of chronic or recovered stages of HBV infection. The use of other viral markers such as HBsAg, anti-HBs, and anti-HBc total to differentiate acute from chronic hepatitis B is inconclusive because most of these markers are alsoseen in chronic infection.

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# Specimen Requirements

### Specimen

Serum **or** plasma

### Volume

1 mL

### Minimum Volume

0.4 mL (**Note:** This volume does **not** allow for repeat testing.)

### Container

Red-top tube, gel-barrier tube, **or** lavender-top (EDTA) tube

### Collection Instructions

If tube other than a gel-barrier tube is used, transfer separated serum or plasma to a plastic transport tube.

### Stability Requirements

| Temperature | Period |
| --- | --- |
| Room temperature | 14 days |
| Refrigerated | 14 days |
| Frozen | 14 days |
| Freeze/thaw cycles | Stable x3 |

### Reference Range

Negative

## Storage Instructions

Room temperature

## Causes for Rejection

Non-EDTA plasma specimen; PST gel-barrier tubes

# References


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Schillie S, Vellozzi C, Reingold A, et al. Prevention of Hepatitis B Virus Infection in the United States. Recommendations of the Advisory Committee on Immunization Practices. *MMWR Recomm Rep.* 2018 Jan 12;67(1):1-31. PubMed 29939980 (http://www.ncbi.nlm.nih.gov/pubmed/29939980)

Terrault NA, Lok ASF, McMahon BJ, et al. Update on Prevention, Diagnosis, and Treatment of Chronic Hepatitis B: AASLD 2018 Hepatitis B Guidance. *Hepatology.* 2018 Apr;67(4):1560-1599. PubMed 29405329 (http://www.ncbi.nlm.nih.gov/pubmed/29405329)

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